

Decarbonising AI-Assisted Software

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University of Salerno
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ITEA GENIUS



**Carbon Footprint of
LLM-Assisted Coding**



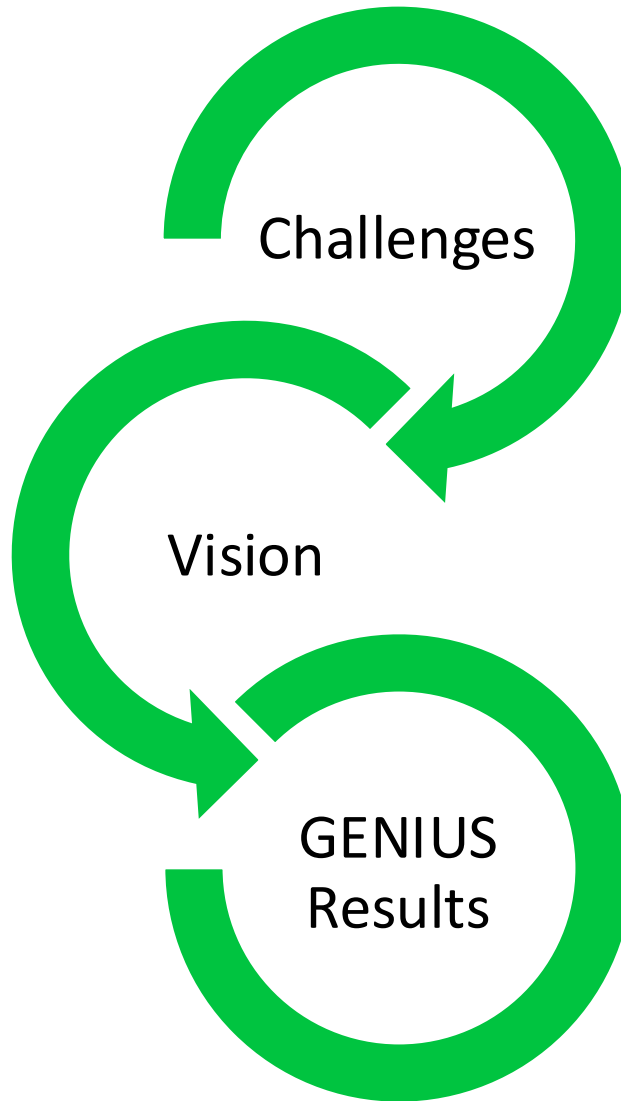
ITEA GreenCode

The Future of Generative AI in Software Engineering: A Vision from Industry and Academia in the European GENIUS Project

Robin Gröpler, Steffen Klepke, Jack Johns, Andreas Dreschinski, Klaus Schmid, Benedikt Dornauer, Eray Tüzün, Joost Noppen, Mohammad Reza Mousavi, Yongjian Tang, Johannes Viehmann, Selin Şirin Aslangül, Beum Seuk Lee, Adam Ziolkowski, Eric Zie



Outline



Challenges



- **Hallucination** and Limited **Reasoning** and Issues with **Structured Output**:
 - Confident but incorrect or unverifiable outputs
 - Lack of semantic understanding
 - Examples: use of deprecated libraries
- **Context** unawareness:
 - Unaware of specific human and company contexts
 - Examples: user expertise, specific requirements, architectural constraints



Challenges



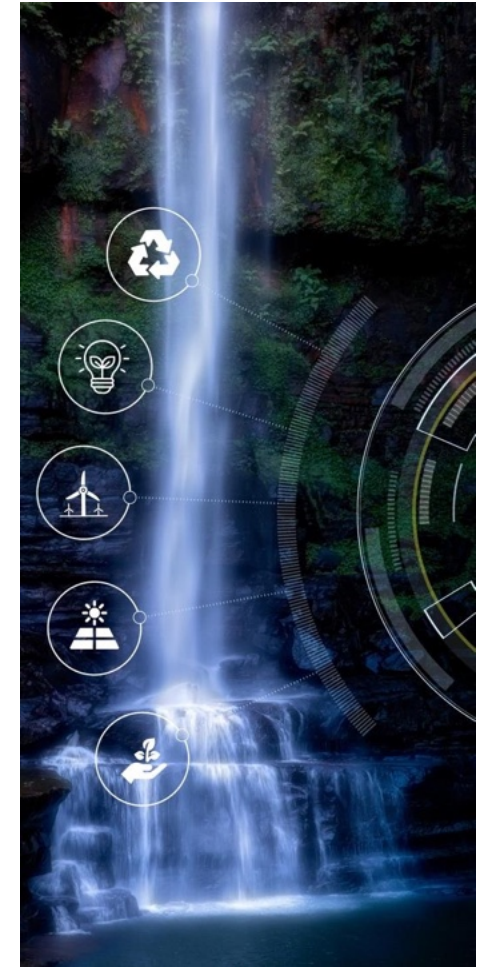
- **Security and Privacy:**
 - Lack of benchmarks and dynamic analysis tools
 - Language dependencies; challenges with low-resource languages
- **Bias:**
 - Well-known fairness issues
 - Understudies implications for code



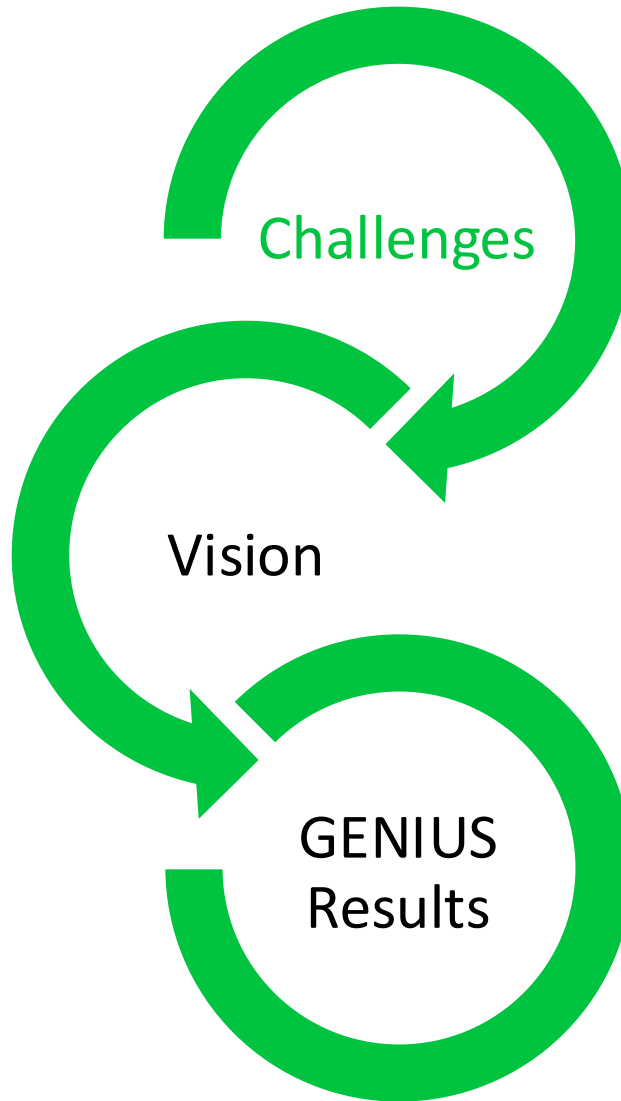
Challenges



- **Sustainability :**
 - No transparent end-to-end models of impact
 - Understudies patters of sustainable Gen-AI-assisted SE
- **Long-term Impacts on SE and New Roles:**
 - Need for new roles and expertise
 - New processes and workflows



Outline



Vision



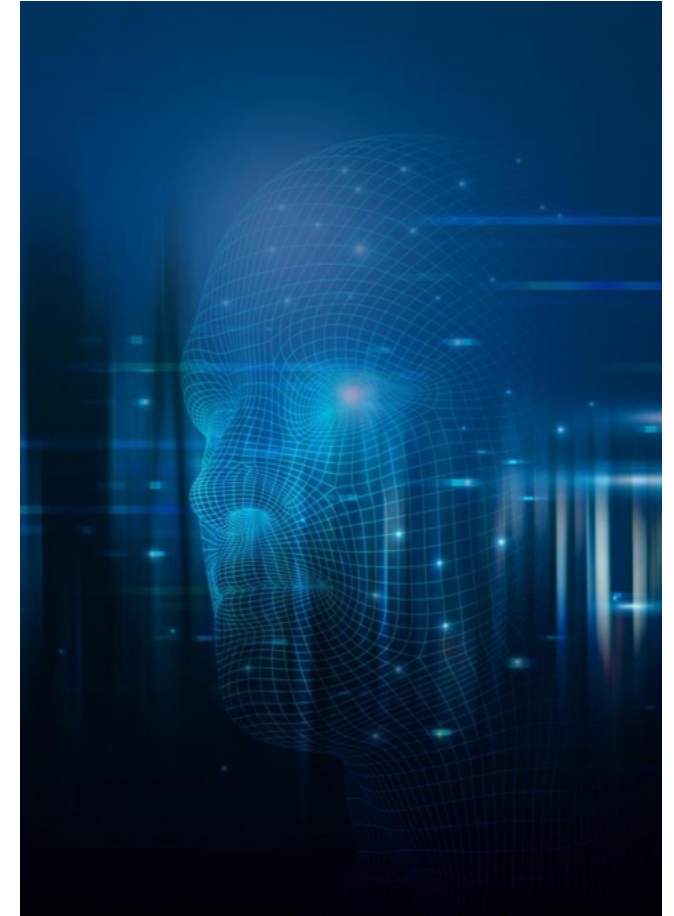
- **Self-* Future:**
 - Emphasis of sustainable minimalism
 - Focus on validation and verification
- **Multi-Modal Human-AI Collaboration:**
 - Emerging interaction patterns
 - New interfaces
 - Maturing knowledge and data management



Vision



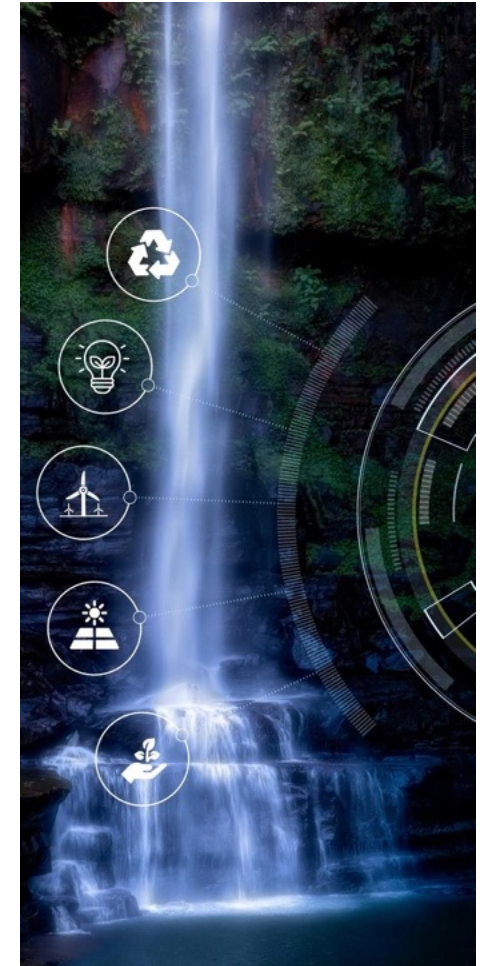
- Improved **trustworthiness**:
 - Extended reasoning interfaces
 - Chain-of-thought models and monitors
- Embedded **Security and Privacy**
 - Structured defenses
 - Privacy-preserving mechanisms



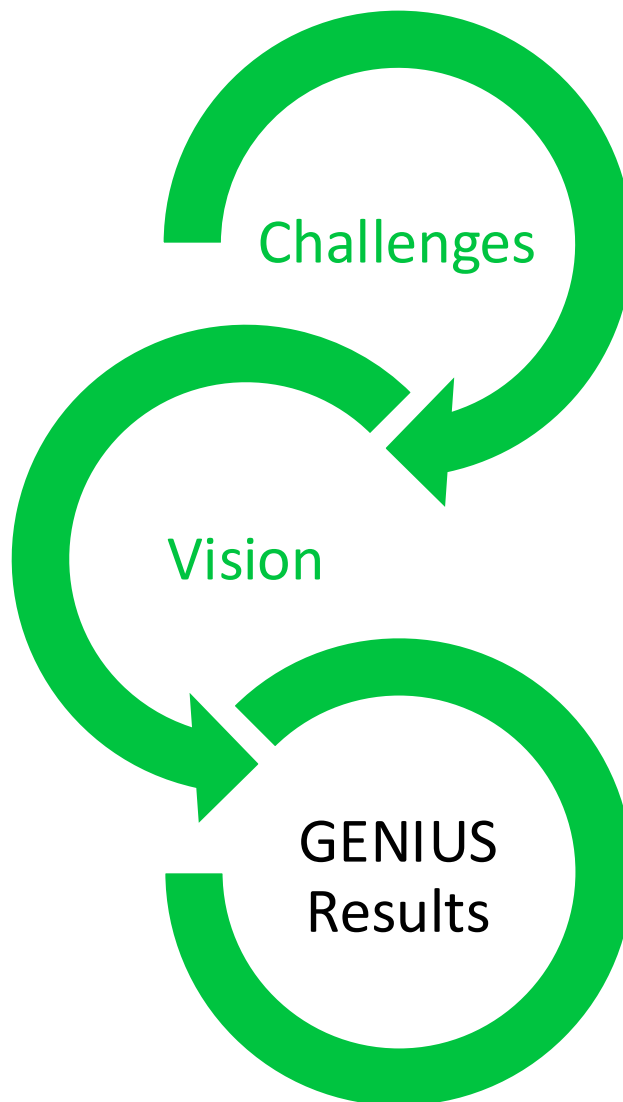
Vision



- **Sustainability :**
 - First-class requirement
 - Designing sustainable processes and patterns
- **New roles and skills**



Outline



GENIUS Results



Overview of the GENIUS Project

- GENIUS is a collaborative European R&D initiative with over 30 industrial and academic partners.
- The project addresses both horizontal and domain-specific challenges in applying GenAI across the SDLC.
- It aligns with EU values and regulations, emphasizing transparency, sovereignty, and trustworthiness.

Use Cases as Drivers for Industrial Relevance and Evaluation

- GENIUS includes 14 industry-defined use cases to drive development and evaluate real-world applicability.
- Use cases span sectors like manufacturing, telecom, automotive, and healthcare, ensuring broad relevance.
- These cases guide the iterative refinement and industrial validation of the developed GenAI tools.



GENIUS Results



Technological Innovations Addressing Core Challenges

- GENIUS is developing context-aware artifact generation pipelines that support traceability and correctness.
- Semantic knowledge management tools like GraphRAG enable effective reuse of past decisions and artifacts.
- Technical safeguards ensure secure, private, and verifiable AI integration in industrial settings.

Expected Outcomes and Impact

- The project aims to deliver production-ready tools, evaluation frameworks, and open-source components.
- Outcomes will reduce development time, improve software quality, and increase organizational agility.
- GENIUS intends to establish GenAI as an integrated, accountable collaborator in software development.



GENIUS Results



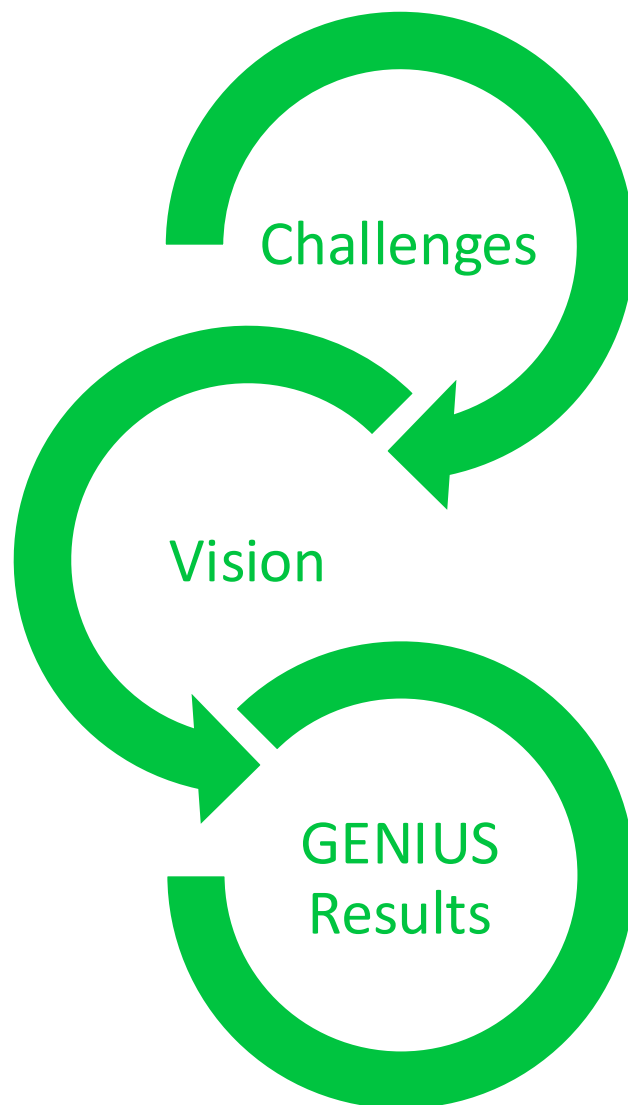
- Sustainability and Environmental Impact:
 - Models of environmental impact
 - Emerging patterns of sustainable LLM-assisted SE



See the next part for more information!



Outline





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**Carbon Footprint of
LLM-Assisted Coding**



ITEA GreenCode

Comparative Analysis of Carbon Footprint in Manual vs. LLM-Assisted Code Development

Authors: Kuen Sum Cheung (King's College London),
Mayuri Kaul (Go Code Green),
Gunel Jahangirova (King's College London),
Mohammad Reza Mousavi (King's College London), Eric Zie (CR&A)

ResponsibleSE 2025, 26th June, Trondheim, Norway



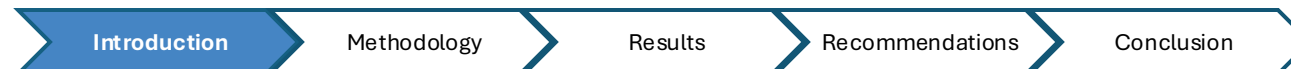
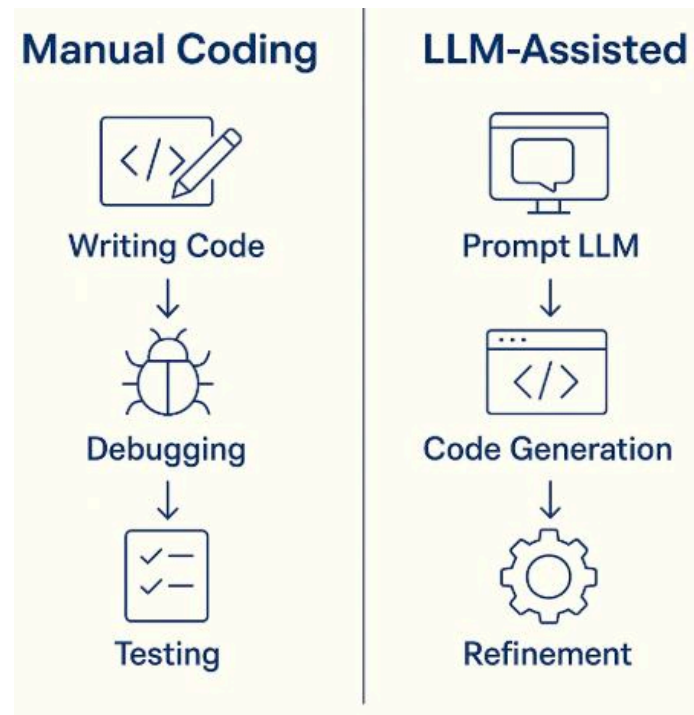
Key Message

LLM-Assisted Coding Results in Higher Carbon Emissions Than Manual Coding



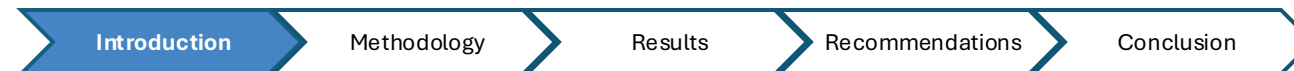
Manual vs LLM-Assisted Coding

- We will explain what these terms mean and the context within our study.



Carbon Emissions

- Metric: carbon emissions (CO₂) for environmental impact of code development
- Methodology: experiment comparing carbon emission of manual and LLM-assisted coding.



Codeforces

Codeforces is an online programming platform. The image on the left is an example of the type of questions asked on the platform.

A. Trippi Troppi

time limit per test: 1 second

memory limit per test: 256 megabytes

Trippi Troppi resides in a strange world. The ancient name of each country consists of three strings. The first letter of each string is concatenated to form the country's modern name.

Given the country's ancient name, please output the modern name.

Input

The first line contains an integer t – the number of independent test cases ($1 \leq t \leq 100$).

The following t lines each contain three space-separated strings. Each string has a length of no more than 10, and contains only lowercase Latin characters.

Output

For each test case, output the string formed by concatenating the first letter of each word.

Example

input

```
7
united states america
oh my god
i cant lie
binary indexed tree
believe in yourself
skibidi slay sigma
god bless america
```

output

```
usa
omg
icl
bit
biy
sss
gba
```

Introduction

Methodology

Results

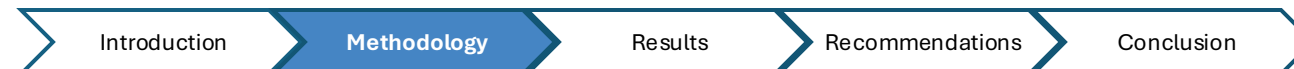
Recommendations

Conclusion



Task Complexity

#	Who	=	Penalty	*	<u>A</u>	<u>B</u>	<u>C</u>	<u>D</u>	<u>E</u>	<u>F</u>	<u>G</u>	<u>H</u>
1	woralem23 [#]	8	97		+ 00:01	+ 00:02	+ 00:03	+1 00:06	+2 00:10	+ 00:13	+ 00:15	+ 00:17
1	 *TKT_YI	8	97		+ 00:02	+ 00:04	+ 00:06	+ 00:08	+ 00:10	+ 00:17	+ 00:22	+ 00:28
3	 rgnerdplayer [#]	8	99		+ 00:00	+ 00:01	+ 00:03	+ 00:07	+ 00:13	+ 00:16	+ 00:22	+ 00:37
4	 *neal	8	101		+ 00:00	+ 00:01	+ 00:03	+ 00:05	+ 00:07	+ 00:19	+ 00:37	+ 00:29
5	*nifeshe	8	102		+ 00:01	+ 00:03	+ 00:05	+ 00:08	+ 00:10	+ 00:14	+ 00:23	+ 00:38
6	 kevinyang [#]	8	104		+ 00:00	+ 00:02	+ 00:05	+ 00:10	+ 00:07	+ 00:14	+ 00:28	+ 00:38
7	 *Hamed_Ghaffari	8	109		+ 00:00	+ 00:08	+ 00:06	+ 00:12	+ 00:14	+ 00:18	+ 00:23	+ 00:28
8	 *abc864197532	8	111		+ 00:00	+ 00:02	+ 00:03	+ 00:06	+ 00:08	+1 00:39	+ 00:17	+ 00:26



CodeCarbon



```
[codecarbon INFO @ 11:45:35] Energy consumed for RAM : 0.000008 kWh. RAM Power : 3.0 W
[codecarbon INFO @ 11:45:36] Energy consumed for all CPUs : 0.000002 kWh. Total CPU Power : 0.87010000000000
881 W
[codecarbon INFO @ 11:45:39] Energy consumed for all GPUs : 0.000000 kWh. Total GPU Power : 0.0049 W
[codecarbon INFO @ 11:45:41] 0.000011 kWh of electricity used since the beginning.
[codecarbon INFO @ 11:45:45] Energy consumed for RAM : 0.000011 kWh. RAM Power : 3.0 W
[codecarbon INFO @ 11:45:46] Energy consumed for all CPUs : 0.000003 kWh. Total CPU Power : 0.89420000000000
881 W
[codecarbon INFO @ 11:45:49] Energy consumed for all GPUs : 0.000000 kWh. Total GPU Power : 0.01 W
[codecarbon INFO @ 11:45:51] 0.000015 kWh of electricity used since the beginning.
[codecarbon INFO @ 11:45:55] Energy consumed for RAM : 0.000014 kWh. RAM Power : 3.0 W
[codecarbon INFO @ 11:45:56] Energy consumed for all CPUs : 0.000005 kWh. Total CPU Power : 1.25660000000000
884 W
[codecarbon INFO @ 11:45:59] Energy consumed for all GPUs : 0.000000 kWh. Total GPU Power : 0.0066 W
[codecarbon INFO @ 11:46:01] 0.000019 kWh of electricity used since the beginning.
```

Introduction

Methodology

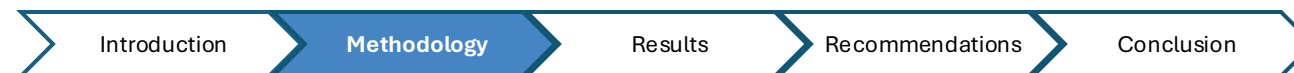
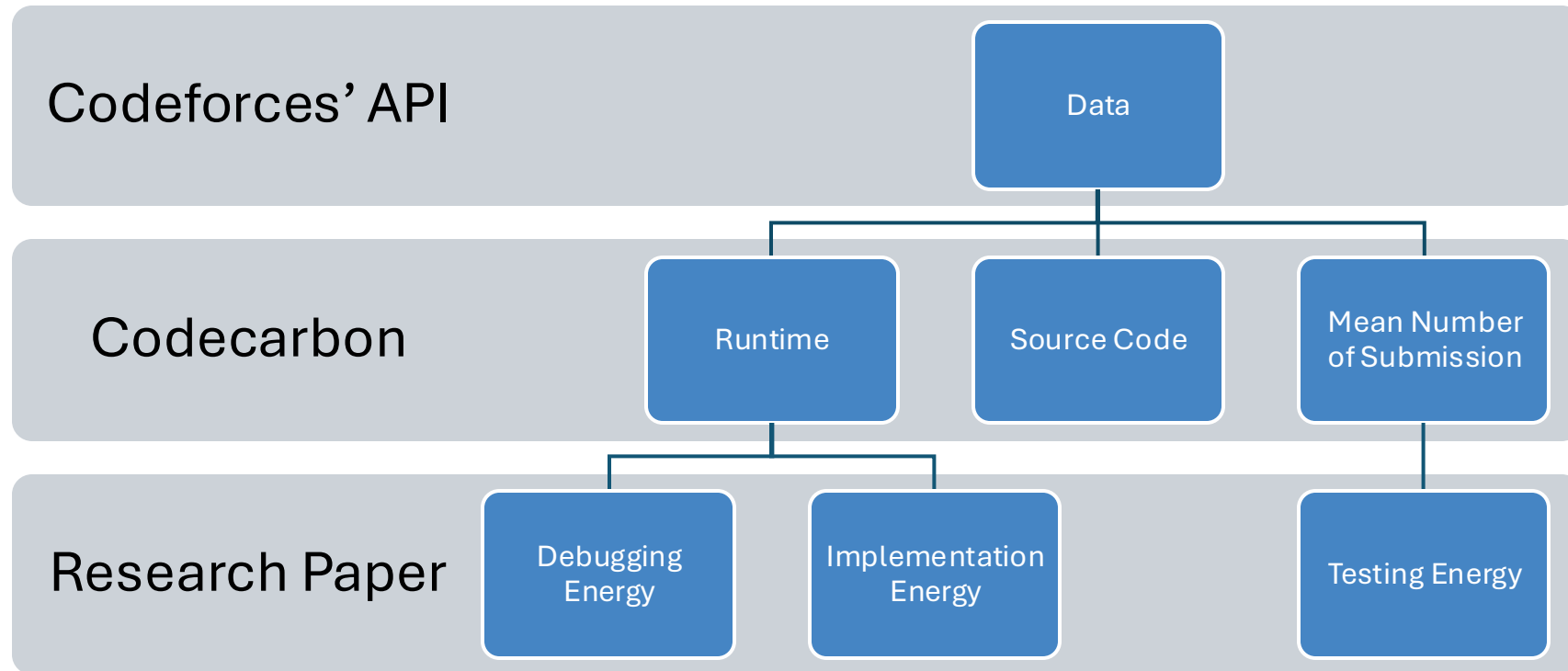
Results

Recommendations

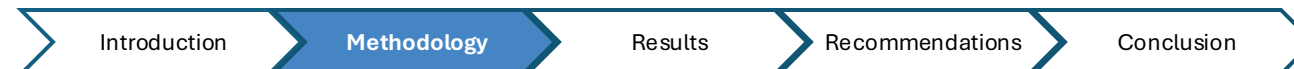
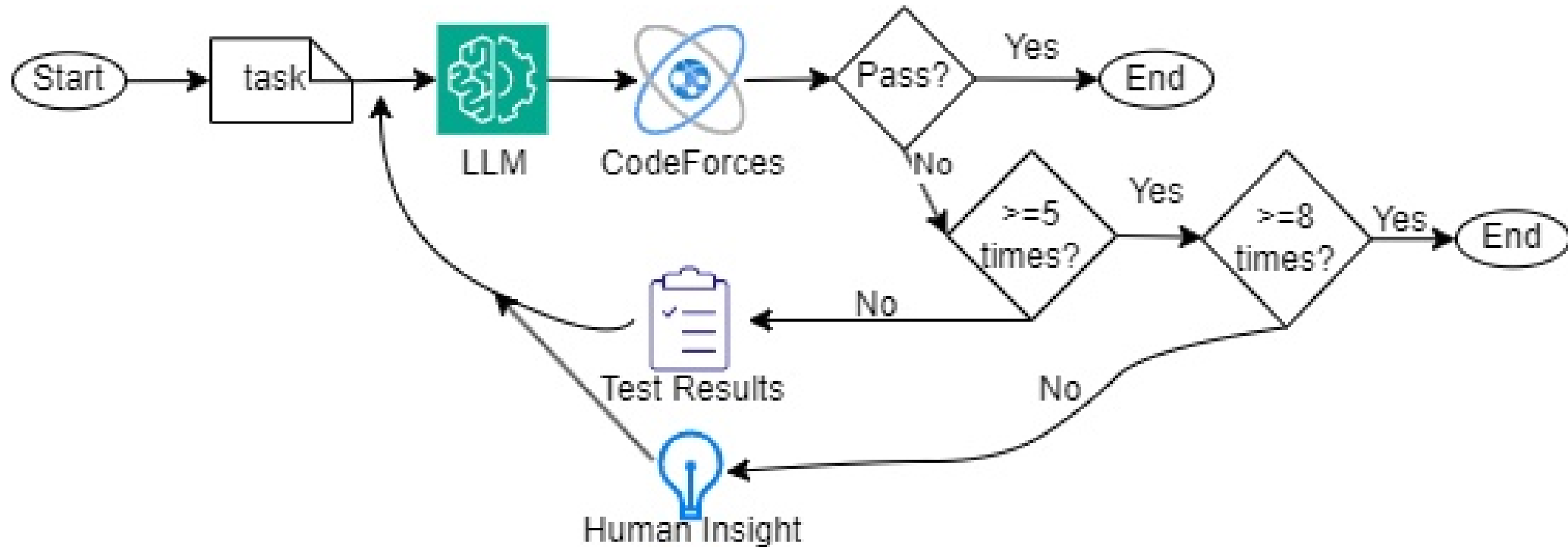
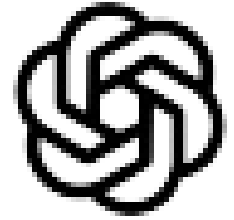
Conclusion



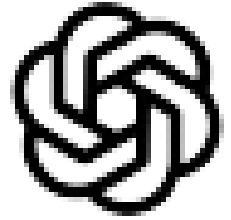
Manual Coding: Methodology



LLM-Assisted Coding: Methodology



Human Insight



2094E - Boneca Ambalabu

Problem Credits: **Proof_by_QED**

▼ Hint 1

Consider each bit independently.

▶ Hint 2

▼ Solution

Here, it suffices to consider each bit independently, because addition is both commutative and associative. For $0 \leq i < 30$, let cnt_i denote the number of elements of a which have the i -th bit set (where bits are indexed from the least significant bit being the 0-th bit); note that these can be found in $O(30n)$ time (more generally, we need $O(\log \max a_i)$ operations per element).

Now, suppose we choose a particular k . Then we can find the sum $(a_k \oplus a_1) + (a_k \oplus a_2) \cdots + (a_k \oplus a_n)$ as follows: for a given bit position i , the number of elements a_j such that $a_k \oplus a_j$ has the i -th bit set will be cnt_i if the i -th bit of a_k is not set, and $n - cnt_i$ if the i -th bit of a_k is set. So we can simply add $cnt_i \cdot 2^i$ to our sum if the i -th bit of a_k is not set, and add $(n - cnt_i) \cdot 2^i$ to our sum if the i -th bit of a_k is set.

This allows us to compute $(a_k \oplus a_1) + (a_k \oplus a_2) \cdots + (a_k \oplus a_n)$ for any k in $O(30)$ time, so we can now compute this sum for all k in $O(30n)$ time and output the maximum.

Introduction

Methodology

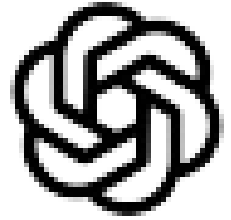
Results

Recommendations

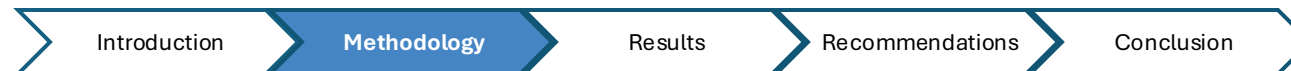
Conclusion



Queries to Energy



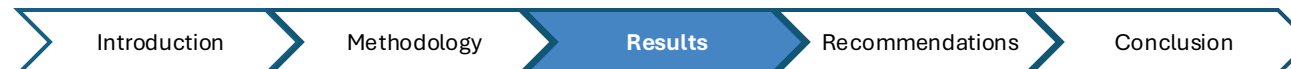
- GPT4
- 0.0022 kwh per inference
- Training cost: 50 Gwh
- Total number of queries: 5.68 Giga queries
- Inference + training = 0.011 kwh per query



Results (RQ1)

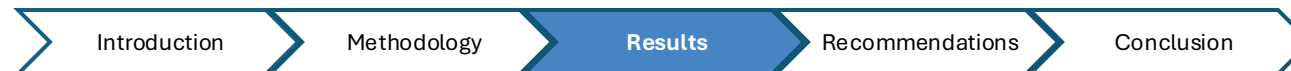
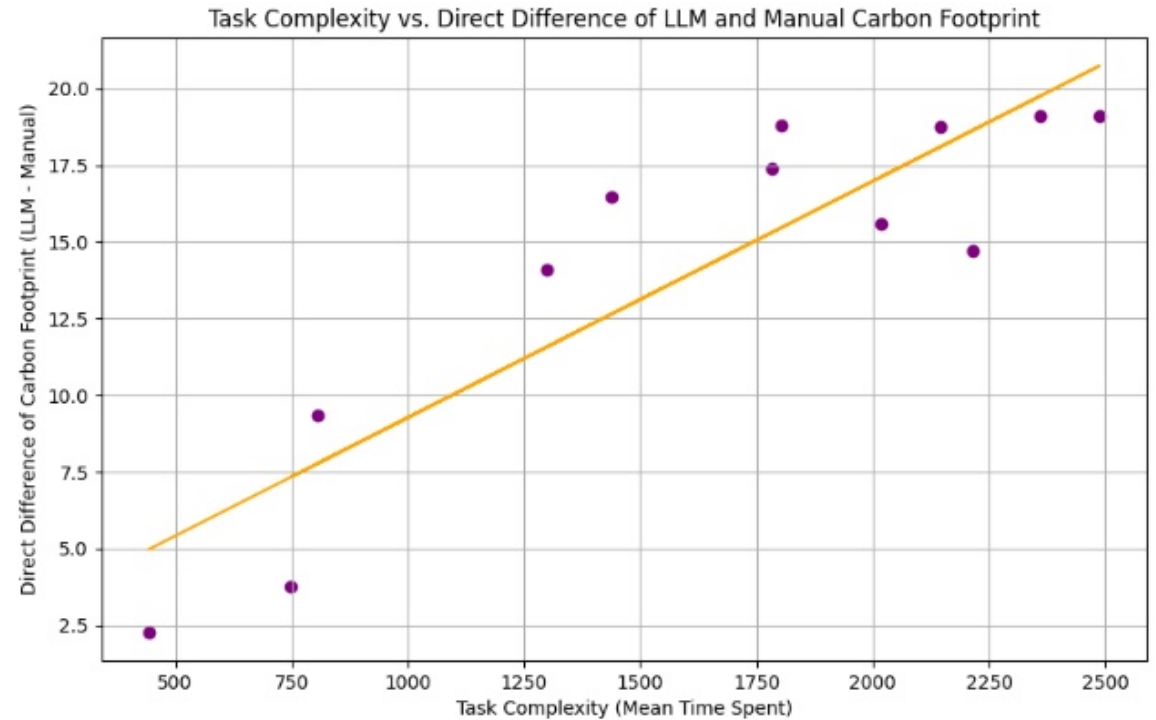
- LLM-assisted software development leads to a significantly higher carbon footprint compared to manual software development.

	Contest 1983					Contest 1984				Contest 1994		
	A	B	C	D	E	A	B	C	D	A	B	C
Manual CF (g)	0.120	0.600	0.640	0.547	0.676	0.218	0.389	0.489	0.582	0.203	0.352	0.483
LLM CF (g)	2.39	15.32	19.77	16.15	19.81	9.55	16.84	19.29	19.35	3.99	14.44	17.86
ratio	19.92	25.53	30.89	29.52	29.30	43.81	43.29	39.45	33.25	19.66	41.02	36.98
Mean: 32.72						Standard deviation: 8.41						



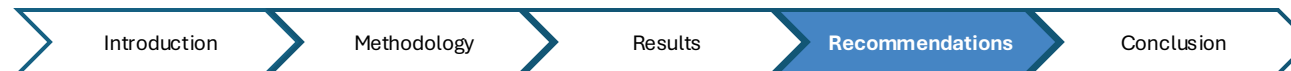
Results (RQ2)

Significant **correlation** between:
task complexity and
difference in carbon footprint

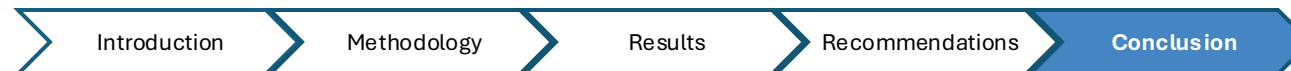
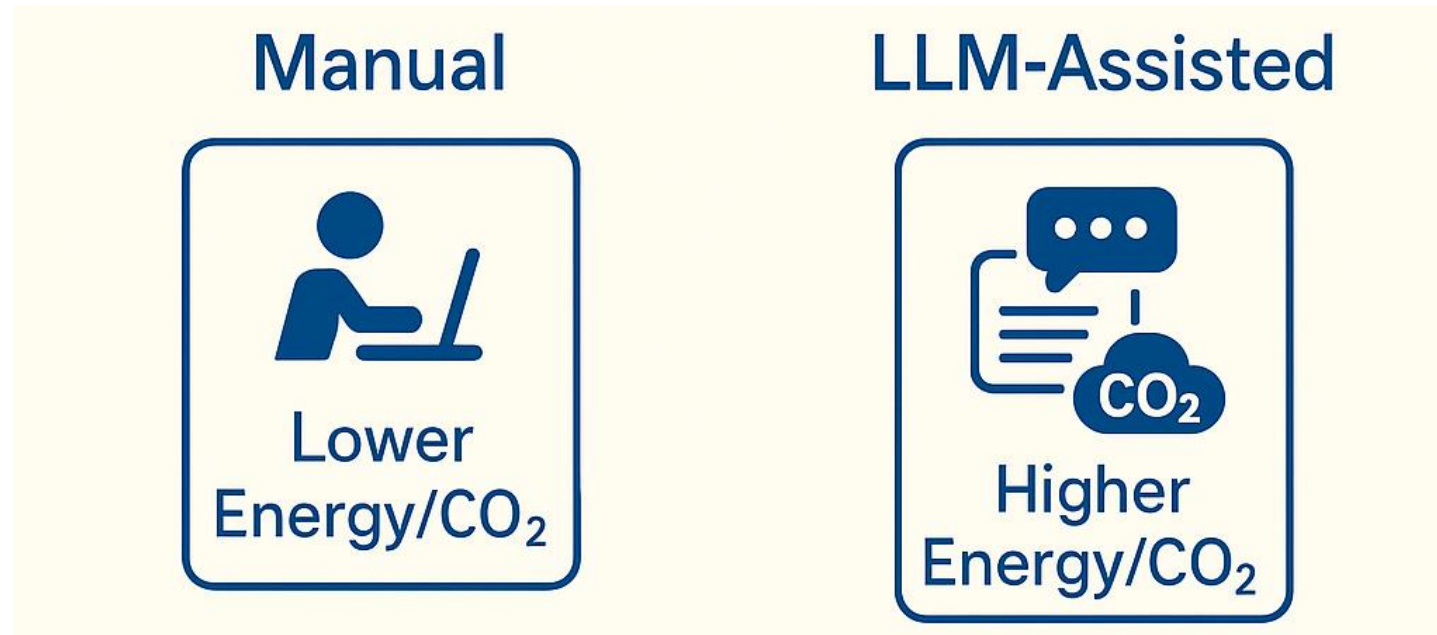


Recommendations

- Assess Task Complexity Before Using LLMs
- Leverage LLMs for Simple Tasks
- Incorporate Human Insight for Medium Tasks
- Decompose Complex Tasks



Conclusions





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Carbon Footprint of
LLM-Assisted Coding 



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Domain-Specific LLMs/VLMs

Autonomous Vehicles Domain

Gao et al. Foundation Models in Autonomous Driving: A Survey on Scenario Generation and Scenario Analysis. Jun 2025.



Domain-Specific LLMs/VLMs

Realism of Scenarios

Q. Song, A. Bensoussan, and M.R. Mousavi.
Synthetic vs. Real: An Analysis of Critical
Scenarios for Autonomous Vehicle Testing.



Domain-Specific LLMs/VLMs

Augmentation

- Using VLMs, diffusion models
- Causal analysis

Domain-Specific LLMs/VLMs

Sustainability

- Energy Profiling:
 - Design-time
 - Runtime
- LLM-assisted refactoring for sustainability



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Carbon Footprint of
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Thank you very much!

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